

Statement of Work

400G Optical Network Upgrade Installation and Commissioning Services

- 1.0 OVERVIEW:** The 633rd Communications Squadron (633 CS) at Langley Air Force Base, Virginia, has a need to upgrade its core optical transport network. This project requires professional services to install and commission Government-Furnished Equipment (GFE), specifically Cisco Network Convergence System 2006 Series (NCS2K) Dense Wavelength Division Multiplexing (DWDM) nodes and 400G Transponder Modules, into the existing network infrastructure to complete this critical modernization effort.
- 2.0 SCOPE:** The scope includes all tasks from initial site preparation through final system testing and delivery of documentation. The outcome shall be a fully operational, secure, and integrated 400Gbps optical network capability across four (4) specified locations on Langley AFB, integrated into the existing eleven (11)-site network environment. The resulting network shall be agile and scalable to meet critical mission demands.
- 3.0 SPECIFIC REQUIREMENT:** The contractor shall provide all project management, personnel, labor, equipment, tools, and technical expertise necessary to perform the installation, configuration, testing, and integration of government-furnished 400G optical networking equipment.
- 3.1. Professional Installation Services:** The contractor shall perform the physical installation of the government-furnished Cisco NCS2K optical nodes and 400G Transponder Modules at the four (4) specified locations. This includes, but is not limited to, the racking, powering, cabling, and labeling of all equipment in accordance with all applicable industry best practices and manufacturer guidelines.
- 3.2. System Commissioning and Integration:** The contractor shall configure the newly installed equipment to bring the 400G circuits into operational status. This shall include loading and verifying software, performing system turn-up and testing, performing database backups, and integrating the new nodes into the existing network management system to ensure full visibility and control.
- 3.3. Downtime and Outage Coordination:** The contractor shall work directly with government points of contact (POCs) to schedule all network downtime and outages. The contractor shall be flexible when failing over to the live network to ensure no impact on critical missions, such as those of the Intelligence Community (IC).
- 4.0 CONTRACTOR PERSONNEL AND QUALIFICATIONS:**
- 4.1. Project Management:** The contractor shall provide comprehensive project management to ensure the project is completed on schedule and within scope. The Contractor shall provide a Project Manager (PM) and alternate(s) responsible for contract performance, coordination with government POCs, management of all on-site activities, and providing regular status updates.
- 4.2. Expert Technical Personnel:** The contractor shall provide certified and experienced engineers and technicians with demonstrated expertise in deploying and commissioning Cisco optical networking platforms, specifically the NCS 2000 series. Personnel shall be capable of working independently and troubleshooting complex integration challenges in a live network environment.
- 5.0 SECURITY:**

5.1 Security Clearance: Contractor personnel assigned to this project shall possess a minimum of a Secret security clearance prior to the contract start date.

5.2 Security Compliance: All installation, configuration, and integration activities shall be performed in strict adherence to all applicable Department of Defense (DoD), U.S. Air Force, and local security standards, policies, and regulations.

5.3 Base and Facility Access: The contractor shall ensure that each employee obtains all necessary passes, identification, and credentials required for access to the installation and secure work areas. The contractor's personnel shall be able to obtain the correct credentials to enter/access any secure building/areas.

6.0 HOURS OF OPERATIONS: The normal hours of operation are 0730 to 1630 Eastern Standard Time (EST), Monday through Friday, excluding Federal Holidays. The contractor shall not be expected to be on-site on federal holidays unless scheduled in advance. The contractor shall be available to perform work outside of normal duty hours, including nights and possible weekends, as required to support mission-critical activities. All work performed outside normal hours shall be scheduled and coordinated with the government POC in advance.

7.0 GOVERNMENT-FURNISHED ITEMS (GFI) AND SERVICES: The government will provide the following at no cost to the contractor:

7.1 Government-Furnished Equipment (GFE): All required networking hardware as specified below. The contractor shall be responsible for inventory and management of all GFE upon receipt.

Qty	Part Number	Description
7	NCS2K-400G-XP	400G CFP2 MR Xponder
14	ONS-CFP2-WDM	100G QPSK / 200G 16-QAM - WDM CFP2 Pluggable
28	ONS-QSFP-4X10-MLR	4x10Gbps Multi-rate QSFP+, LR
7	15216-HD-EXT-PNL	Mechanical Frame - 4 slots - 1 RU
14	NCS2K-MF-8X10G-FO	2x4x10G QSFP+ to 10G Fan Out
28	ONS-12MPO-MPO-8	Multi-fiber patch cord - MPO to MPO 12 fibers - 8m
14	15454-M-USBCBL	USB cable for passive devices
28	15454-LC-LC-2	Fiber patch cord - LC to LC - 2m
35	C9300X-24Y-A	Catalyst 9300X 24x25G Fiber Ports, modular uplink Switch
35	PWR-C1-715WAC-P/2	715W AC 80+ platinum Config 1 Secondary Power Supply
35	C9300X-NM-8Y	Catalyst 9300 8 x 10G/25G Network Module SFP+/SFP28

7.2 Facility Access: Access to all required installation locations.

7.3 Security Escorts: 633 CS will provide escorts to facilities and Communications rooms as required.

7.4 Workspace: The government will provide contractor office/desk space(s) as available for the duration of on-site activities.

8.0 DELIVERABLES AND DOCUMENTATION: The contractor shall provide the following deliverables in electronic (PDF) format:

Deliverable	Due Date
Low-Level Design (LLD)	10 business days after kickoff meeting
Implementation Plan	10 business days after LLD approval
Final As-Built Documentation	15 business days after project completion

The Final As-Built Documentation shall be comprehensive, including system diagrams, equipment specifications, software manuals, network diagrams, maintenance procedures, final configurations, how-to manuals, and a record of all labeling.

9.0 PLACE OF PERFORMANCE: All services shall be performed on-site at Langley Air Force Base, Virginia. The specific building locations will be provided to the contractor upon award.

10.0 OPTIONAL REQUIREMENT:

10.1. On-Site Training: Upon request by the government, the contractor shall provide an informal, hands-on 400G Optical Network Operation, Administration, Maintenance, and Provisioning (OAM&P) training workshop for up to twelve (12) government personnel. This optional task, if exercised, shall be scheduled by the government and conducted either during the 90-day period of performance or within 30 days following project completion.